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CSA0609- Design and Analysis of Algorithms

Bubble Sort with Big-O Analysis

AIM

To implement the Bubble Sort algorithm in Python and analyze its time and space complexity.

ALGORITHM

1. Start.
2. Compare adjacent elements of the array.
3. If the first element is greater than the second, swap them.
4. Repeat the comparison for all adjacent elements.
5. Repeat the passes until the array is sorted.
6. Display the sorted array.
7. Stop.

PSEUDOCODE

```
BubbleSort(A)
for i = 0 to n-2
  for j = 0 to n-i-2
    if A[j] > A[j+1]
      swap A[j], A[j+1]
    end if
  end for
end for
```

PYTHON CODE

main.py Visualize Run ▶

```python
def bubble_sort(A):
    n = len(A)

    for i in range(n - 1):
        for j in range(n - i - 1):
            if A[j] > A[j + 1]:
                A[j], A[j + 1] = A[j + 1], A[j]

A = [64, 34, 25, 12, 22, 11, 90]
bubble_sort(A)

print("Sorted Array:", A)
```

OUTPUT

Output

```
Sorted Array: [11, 12, 22, 25, 34, 64, 90]
```

COMPLEXITY ANALYSIS

Let n be the number of elements.

- Best Case: $O(n^2)$
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$
- Space Complexity: $O(1)$

Big-O Time Complexity: $O(n^2)$

RESULT

Thus, the Bubble Sort algorithm was successfully implemented in Python, and its time complexity was analyzed as $O(n^2)$ with $O(1)$ space complexity.